



Neurotransmitters

اسم الطالب : علي امير يحيى الشهاري

اسم المشرف/ د. علي الحاج



جامعة العلوم والتكنولوجيا
University of Science & Technology

1. Introduction

The central nervous system (CNS) processes information from and delivers information to the peripheral nervous system through signal conduction from one neuron to another via synapses. Thus, through synaptic transmission (also known as neurotransmission), CNS can control smooth, skeletal, and cardiac muscles, bodily secretions, and organ functions.

An essential role in information transmission throughout the CNS and peripheral nervous system is played by neurotransmitters (NTs), which are endogenous chemical messengers that carry and amplify nerve-to-nerve signaling or signals between nerves and other cell types. These small molecules are crucial for communicating sensory, motor, and integrative neuronal messages, affecting many functions, such as emotions, thoughts, memories, movements, and sleep patterns. These chemicals play essential roles in the functioning of the brain, being fundamental regulators of neuronal growth, differentiation, and survival. Consequently, abnormal levels of NTs are reflected in dysregulation of brain functions, leading to various physical, psychotic, and neurodegenerative diseases [2,3,4,5,6,7].

In this context, a deeper understanding of NTs' roles and actions is mandatory for better assessing some of the most severe neurological disorders. Therefore, this review aims to shed some light on the various known canonical and noncanonical NTs, emphasizing their involvement in different relevant diseases. Moreover, novel NT detection methods are presented, followed by several modulation strategies for treating associated neurological conditions.

2. Neurotransmitters

Neurotransmitters (NTs) are endogenous chemical messengers that amplify, transmit and convert signals between nerve cells and from nerves to other cell types. They play a central role in brain function, behavior and cognition, and are essential in the nervous system for linking sensory input, motor output and integrative neural processes.

According to the authors, more than 200 distinct neurotransmitter molecules have been identified since the first recognition in 1921, yet the exact total remains unknown as new bio-active molecules continue to be discovered.

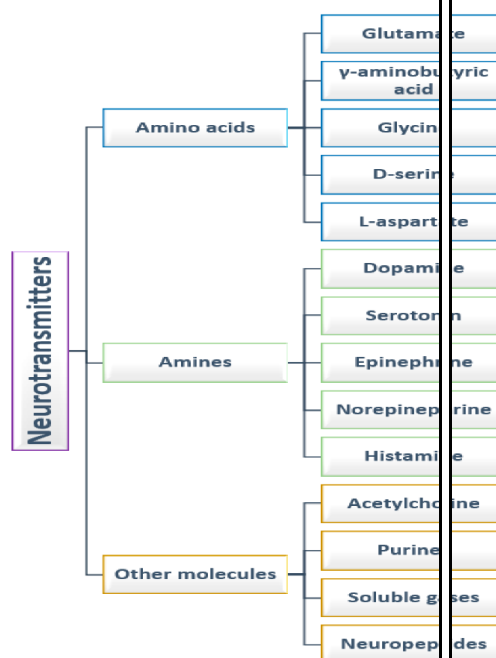


Figure 1 of the article illustrates a classification of neurotransmitters into the main categories (“canonical” and “non-canonical”).

Classification of Neurotransmitters:

The review distinguishes two broad classes of NTs:

Figure 1: Canonical and Non-canonical

1. **Canonical NTs:** These are well-established chemical messengers. They include amino acid neurotransmitters (e.g., glutamate, γ -aminobutyric acid [GABA], glycine, D-serine, L-aspartate), monoamines (e.g., dopamine, serotonin, norepinephrine, epinephrine, histamine), acetylcholine, purines (e.g., ATP), soluble gases (e.g., nitric oxide [NO], hydrogen sulfide [H_2S], carbon monoxide [CO]) and neuropeptides (e.g., neuropeptide Y, substance P, vasopressin, β -endorphin).

Amino acids: For example, glutamate is the major excitatory NT in the central nervous system; excessive glutamate signalling (excitotoxicity) is implicated in multiple neurodegenerative and neurological disorders. GABA is the main inhibitory NT in the adult brain, synthesized via glutamate decarboxylase; its dysfunction is linked to epilepsy, autism, schizophrenia, sleep and motor disorders. Glycine acts as inhibitory transmitter in the spinal cord and brainstem and as a co-agonist of NMDA receptors. D-serine originates from astrocytic conversion of L-serine and serves as a modulator at NMDA receptors. L-aspartate has a more contested role but is present in vesicular pools and may act along with glutamate or GABA.

Monoamines: These include dopamine, serotonin, norepinephrine/epinephrine, histamine, etc. They are produced in discrete brain nuclei (e.g., substantia nigra/ventral tegmental area for dopamine; raphe nuclei for serotonin; locus coeruleus for norepinephrine) and are removed by specific enzymes (e.g., monoamine oxidase, catechol-O-methyltransferase) or by re-uptake transporters. Imbalances in their levels and signalling are tied to Parkinson’s disease, Alzheimer’s disease, Huntington’s disease, schizophrenia, addiction and mood disorders.

Acetylcholine (ACh): It is the first-recognized neurotransmitter, critical in the peripheral nervous system (neuromuscular junctions) and in the central nervous system (attention, learning, memory, voluntary movement, sleep). Degeneration of cholinergic systems is implicated in Alzheimer’s disease, Parkinson’s disease, Huntington’s disease and myasthenia gravis.

Purines, gases and neuropeptides: Purines such as ATP serve dual roles as energy currency and as signalling molecules in synapses; NO, CO and H_2S are gaseous transmitters released by activation of NMDA receptors and act rapidly and diffusely; neuropeptides are abundant in the brain, modulating synaptic activity, plasticity, feeding, circadian rhythm, cardiovascular functions, and are linked with epilepsy and other disorders.

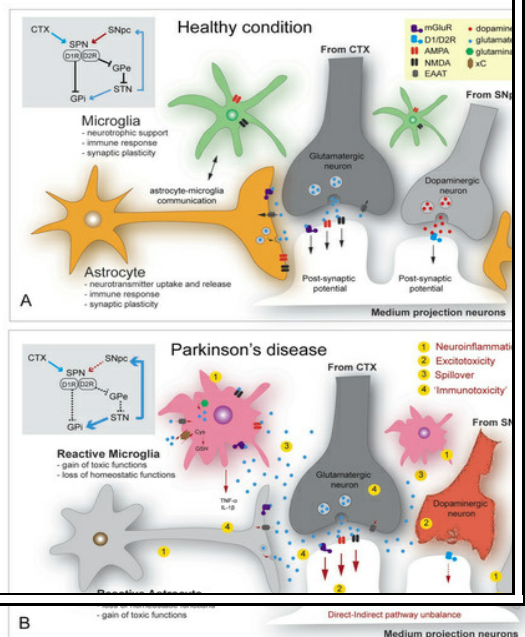
2. Non-Canonical NTs: These include molecules that are more recently proposed or still under debate as true neurotransmitters. Examples include exosomes (neuronal extracellular vesicles carrying proteins, RNAs, receptors and lipids, which influence synaptic plasticity and may act as neuromodulators), steroids (which may act via membrane receptors or transduce rapid non-genomic signalling affecting Ca^{2+} , Src/MAPK, PI3K–AKT pathways) and D-aspartic acid (which is present in nervous tissue, contained in synaptic vesicles and released upon stimulation, and thus a potential transmitter).

3. Roles of Neurotransmitters in Physiology and Disease

Neurotransmitters (NTs) play essential roles in neuronal development, differentiation, synaptic plasticity, and homeostasis in the nervous system. They regulate neuronal growth, shape synaptic connectivity, and maintain the excitatory/inhibitory balance necessary for learning, memory, movement, sleep–wake cycles, and mood regulation. Disruption of this finely tuned NT network impairs neural circuits, contributing to a wide range of neurological and neurodegenerative disorders.

Excitatory neurotransmitters, such as glutamate, can trigger excitotoxicity when excessively released, leading to oxidative stress, synaptic degeneration, and neuronal death. In contrast, deficiency in inhibitory neurotransmitters like GABA can shift the network toward over-activation, producing dysfunction in neuronal circuits. Monoaminergic systems—including dopamine, serotonin, and norepinephrine—modulate mood, motor control, and cognitive functions, with their imbalance linked to depression, schizophrenia, movement disorders, and dementia. Cholinergic deficits affect attention and memory and are strongly associated with Alzheimer's disease. Purinergic, gaseous, and peptide neurotransmitters contribute to neuroinflammation, glial activation, and neurovascular regulation, further influencing disease progression. Figure 2 presents a schematic overview of neurotransmitter-related pathways implicated in major neurological disorders.

Restoring neurotransmitter balance through receptor agonists and antagonists, re-uptake inhibitors, and enzyme modulators represents a cornerstone of current therapeutic strategies for neurological disorders such as Parkinson's disease, depression, and epilepsy. Advances in detection technologies and mapping of neurotransmitter systems (Figure 2) are significantly enhancing our understanding of neurotransmitter



dysfunction in disease and facilitating the development of more targeted and effective treatment approaches.

4. Detection Methods and Technological Advances

Modern techniques for detecting neurotransmitters (NTs) in vitro and in vivo provide crucial insights into their dynamics, concentrations, and spatial distribution. Methods include micro dialysis, fast-scan cyclic voltammetry (FSCV), high-performance liquid chromatography (HPLC) coupled with mass spectrometry, enzyme-based biosensors, and advanced imaging modalities such as positron emission tomography (PET).

Miniaturized nano electrochemical sensors allow high temporal and spatial resolution, which is essential given the rapid kinetics and low concentration of NTs in specific brain regions. These technologies enable real-time monitoring of synaptic events and investigation of NT dysregulation, linking detection capabilities directly to understanding disease mechanisms and developing therapeutic strategies.

5. Modulation Strategies and Therapeutic Applications:

Neurotransmitter systems can be modulated to restore homeostasis and treat neurological disorders. Key strategies include the use of receptor agonists and antagonists, inhibitors of re-uptake transporters, enzyme modulators, gene therapy, and cell-based approaches. Clinically, these approaches are applied as dopamine replacement therapies in Parkinson's disease, selective serotonin re-uptake inhibitors (SSRIs) in depression, and GABAergic modulators in epilepsy. Emerging strategies target purinergic, gaseous, and peptide neurotransmitters to address complex neural dysfunctions. Improved understanding of NT networks, coupled with advanced detection technologies, supports the development of precise therapeutic interventions that effectively restore neurotransmitter balance.

6. Future, Directions and Conclusions:

The landscape of neurotransmitter research continues to evolve, with new NTs and neuromodulators being discovered and mapped with increasing precision. Future research focuses on integrating multi-omics and neuroimaging approaches to comprehensively chart NT networks, developing minimally invasive, high-speed, multiplexed sensors for in vivo NT measurement, and designing personalized therapies based on individual NT profiles. Additionally, understanding the roles of non-canonical NTs will further illuminate mechanisms of neurological and neurodegenerative disorders. Restoring neurotransmitter balance remains central to managing these conditions, and combining advanced detection, modulation strategies, and system-level insights is critical for developing effective, targeted interventions.

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